

A second-order well-balanced reconstruction for the shallow flows with wet/dry fronts

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ABSTRACT

In this paper, we develop a two-dimensional, second-order central-upwind scheme based on the finite volume method for the shallow water equations in the presence of wet/dry fronts. The proposed scheme is positivity-preserving and well-balanced, and remains robust for shallow water flows over complex bathymetry and wet-dry interfaces. We extend existing one-dimensional positivity-preserving and well-balanced schemes to two-dimensional spaces, addressing the challenges posed by partially flooded cells with varying bottom gradients in each direction. A novel draining time step for two-dimensional spaces is introduced to ensure the non-negativity of computed water depths across the entire computational domain. The performance of the proposed scheme is validated through several numerical experiments using both analytical solutions and experimental data, demonstrating its accuracy in capturing wet/dry fronts. The results for the lake-at-rest steady-state case confirm that numerical oscillations near the wet/dry fronts are successfully minimized, maintaining the initial state. Moreover, other examples show reduced numerical oscillations during wave run-up and rundown processes, further confirming the performance of the proposed scheme.

1. Introduction

The shallow water equations (SWEs) are a hyperbolic system of partial differential equations that describe long waves in situations where the horizontal length scale is much greater than the vertical scale (e.g., the flow depth), such as rivers and oceans. Owing to computational efficiency, the SWEs have been widely used in modeling long-wave phenomena including tides, storm surges, tsunamis, and associated transport phenomena [e.g.1–11]. Consequently, their applications are fundamental in coastal hazard prediction and risk management. In their simplest form, the SWEs consist of conservation laws for mass and momentum. In the two-dimensional case, the SWEs are written as

$$\begin{cases} h_t + (hu)_x + (hv)_y = 0, \\ (hu)_t + (hu^2 + \frac{1}{2}gh^2)_x + (huv)_y = -ghB_x, \\ (hv)_t + (huv)_x + (hv^2 + \frac{1}{2}gh^2)_y = -ghB_y, \end{cases} \quad (1)$$

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where subscripts denote partial derivatives with respect to the temporal (t) and spatial variables (x, y). h is the water depth, u and v are the depth-averaged flow velocities in the x and y directions, respectively, B is the bottom elevation, and g is the gravitational acceleration.

A significant challenge in solving the SWEs arises from treating wetting and drying processes, which are crucial for simulating nearshore hydrodynamics including wave run-up and rundown processes. In many applications, when using a coarse grid to compute quasi-steady solutions, small perturbations of steady states can be amplified by the numerical scheme, leading to significant computational errors. Since the SWEs are a system of balance laws, the numerical method must balance the flux and source terms to avoid these artificial errors. In other words, the numerical scheme should preserve the “lake at rest” steady states:

$$w := h + B \equiv \text{Const}, \quad u \equiv v \equiv 0. \quad (2)$$

where $w = h + B$ is the water surface elevation. When $h = 0$, the “lake at rest” state reduces to the “dry lake” steady state, which is defined as

$$h = 0, \quad u = v = 0. \quad (3)$$

The methods that exactly preserve the stationary steady states such as “lake at rest”, “dry lake”, and their combinations are called well-balanced, implying that the flux terms are perfectly balanced with the source terms.

Another difficulty may occur near dry ($h = 0$) or almost dry ($h \approx 0$) regions. In such situations, the water depth may become negative due to numerical oscillations, and then the numerical computations will immediately break down as the eigenvalues of the Jacobian of the SWEs are $u \pm \sqrt{gh}$. Therefore, the numerical scheme needs to ensure the positivity-preserving property that guarantees the non-negativity of the water depth over the entire domain.

In summary, a desirable numerical scheme should possess positivity-preserving and well-balanced properties. There have been many studies devoted to the development of well-balanced positivity-preserving schemes for the SWEs [e.g.12–19]. Kurganov and Petrova [14] developed a robust second-order central-upwind scheme that preserves positivity for shallow water flows over complex bathymetry. This scheme has been widely adopted for shallow water flow applications due to its robustness and accuracy [e.g.20–24]. They adopted a piecewise linear reconstruction method for the conservative variables and a continuous piecewise linear interpolation of the bottom topography. This allows one to guarantee that the reconstructed water surface stays above the discretized bottom topography, which, in turn, ensures the positivity-preserving property of the resulting scheme provided a proper Courant-Friedrichs-Lewy (CFL) condition is satisfied [19]. The well-balanced property is maintained by utilizing an appropriate discretization of the source term [25] for almost dry cells. Recognizing the challenges of unbalanced piecewise linear reconstruction in Kurganov and Petrova’s scheme at wet/dry fronts, Bollermann et al. [16] developed a novel reconstruction method to preserve both the positivity-preserving and well-balanced properties in the vicinity of the wet/dry fronts in one-dimensional space by introducing the concept of partially flooded cells. They also implemented a draining time step that cuts off the outgoing fluxes for partially flooded cells to avoid a possibly severe time step constraint. Horváth et al. [17] further extended the scheme of Bollermann et al. to two dimensions, applying the draining time step subject to the cell-averaged water depth.

This paper introduces a two-dimensional positivity-preserving, well-balanced numerical scheme for the SWEs. Building on the one-dimensional positivity-preserving well-balanced scheme by Bollermann et al. [16], we extend the scheme to two-dimensional spaces. To address inherent issues specific to the two-dimensional geometry of partially flooded cells, we introduce robust criteria for identifying dry cells using cell-averaged variables. Additionally, the draining time step for two-dimensional spaces is implemented to overcome severe time step constraints for partially flooded cells. In the finite-volume formulation, the water depth within a cell is uniquely defined by its cell-averaged value; however, the local water depths defined within the cell may differ between the x - and y -directions in partially flooded cells. Accordingly, the draining time step is computed separately for each direction. This approach guarantees the non-negativity of the computed water depths without necessitating smaller time steps. Several examples with analytical solutions or experimental data were conducted to demonstrate the performance of the proposed scheme.

The remainder of this paper is organized as follows: Section 2 introduces the proposed wet/dry reconstruction scheme for the SWEs, and it also demonstrates the proof that this scheme guarantees the positivity-preserving and well-balanced properties. Section 3 presents the simulation results of several numerical experiments to investigate the performance of the proposed scheme. Finally, Section 4 summarizes the conclusions.

2. Numerical scheme

The two-dimensional shallow water equations in the vector form read:

$$\begin{bmatrix} h \\ hu \\ hv \end{bmatrix}_t + \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \\ huv \end{bmatrix}_x + \begin{bmatrix} hv \\ huv \\ hv^2 + \frac{1}{2}gh^2 \end{bmatrix}_y = \begin{bmatrix} 0 \\ -ghB_x \\ -ghB_y \end{bmatrix}, \quad (4)$$

Using the approach described in [14,25], the system (4) can be reformulated in an equivalent form by redefining the conservation variables as $U := (w, hu, hv)^T$. This can be achieved by substituting h with $w - B$. In this context, it is assumed that B remains constant over time.

2.1. A central-upwind scheme for the shallow water equations

We introduce finite volume cells $I_{i,j} := \left[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}\right] \times \left[y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}\right]$, with uniform grids $x_a := \alpha\Delta x$ and $y_\beta := \beta\Delta y$. The central-upwind semi-discretization of the system (4) reads as

$$\frac{d}{dt} \bar{\mathbf{U}}_{i,j}(t) = -\frac{\mathbf{H}^x_{i+\frac{1}{2},j} - \mathbf{H}^x_{i-\frac{1}{2},j}}{\Delta x} - \frac{\mathbf{H}^y_{i,j+\frac{1}{2}} - \mathbf{H}^y_{i,j-\frac{1}{2}}}{\Delta y} + \bar{\mathbf{S}}_{i,j}(t), \quad (5)$$

where $\bar{\mathbf{U}}_{i,j}(t)$ are the cell averages of the solution $\mathbf{U}$:

$$\bar{\mathbf{U}}_{i,j}(t) \approx \frac{1}{\Delta x \Delta y} \iint_{I_{i,j}} \mathbf{U}(x, y, t) dx dy, \quad (6)$$

$\bar{\mathbf{S}}_{i,j}(t)$ is an appropriate discretization of the cell averages of the source term $\mathbf{S} := (0, -ghB_x, -ghB_y)^T$:

$$\bar{\mathbf{S}}_{i,j}(t) \approx \frac{1}{\Delta x \Delta y} \iint_{I_{i,j}} \mathbf{S}(\mathbf{U}(x, y, t), B(x, y)) dx dy, \quad (7)$$

and the central-upwind numerical fluxes H^x and H^y read as

$$\begin{aligned} \mathbf{H}^x_{i+\frac{1}{2},j} &= \frac{a^+_{i+\frac{1}{2},j} \mathbf{F}\left(\mathbf{U}^-_{i+\frac{1}{2},j}, B\left(x_{i+\frac{1}{2}}, y_j\right)\right) - a^-_{i+\frac{1}{2},j} \mathbf{F}\left(\mathbf{U}^+_{i+\frac{1}{2},j}, B\left(x_{i+\frac{1}{2}}, y_j\right)\right)}{a^+_{i+\frac{1}{2},j} - a^-_{i+\frac{1}{2},j}} \\ &\quad + \frac{a^+_{i+\frac{1}{2},j} a^-_{i+\frac{1}{2},j}}{a^+_{i+\frac{1}{2},j} - a^-_{i+\frac{1}{2},j}} \left[\mathbf{U}^+_{i+\frac{1}{2},j} - \mathbf{U}^-_{i+\frac{1}{2},j} \right], \\ \mathbf{H}^y_{i,j+\frac{1}{2}} &= \frac{b^+_{i,j+\frac{1}{2}} \mathbf{G}\left(\mathbf{U}^-_{i,j+\frac{1}{2}}, B\left(x_i, y_{j+\frac{1}{2}}\right)\right) - b^-_{i,j+\frac{1}{2}} \mathbf{G}\left(\mathbf{U}^+_{i,j+\frac{1}{2}}, B\left(x_i, y_{j+\frac{1}{2}}\right)\right)}{b^+_{i,j+\frac{1}{2}} - b^-_{i,j+\frac{1}{2}}} \\ &\quad + \frac{b^+_{i,j+\frac{1}{2}} b^-_{i,j+\frac{1}{2}}}{b^+_{i,j+\frac{1}{2}} - b^-_{i,j+\frac{1}{2}}} \left[\mathbf{U}^+_{i,j+\frac{1}{2}} - \mathbf{U}^-_{i,j+\frac{1}{2}} \right], \end{aligned} \quad (8)$$

where

$$\begin{aligned} \mathbf{F}(\mathbf{U}, B) &:= \left(hu, \frac{(hu)^2}{w-B} + \frac{1}{2}g(w-B)^2, \frac{(hu)(hv)}{w-B} \right), \\ \mathbf{G}(\mathbf{U}, B) &:= \left(hv, \frac{(hv)^2}{w-B} + \frac{1}{2}g(w-B)^2, \frac{(hu)(hv)}{w-B} \right). \end{aligned} \quad (9)$$

The values $\mathbf{U}^\pm = (u^\pm, h^\pm \cdot u^\pm, h^\pm \cdot v^\pm)$ represent point values at the cell interface of the piecewise linear reconstruction $\tilde{\mathbf{U}} \equiv (\tilde{u}, \tilde{h}u, \tilde{h}v)$,

$$\tilde{\mathbf{U}}(x, y) := \bar{\mathbf{U}}_{i,j} + (\mathbf{U}_x)_{i,j}(x - x_i) + (\mathbf{U}_y)_{i,j}(y - y_j), (x, y) \in I_{i,j}. \quad (10)$$

The numerical derivatives $(U_x)_{i,j}$ and $(U_y)_{i,j}$ are computed using a non-linear limiter. To ensure the non-oscillatory nature of the reconstruction, the generalized minmod limiter is used:

$$\begin{aligned} (U_x)_{i,j} &= \text{minmod}\left(\theta \frac{\bar{\mathbf{U}}_{i,j} - \bar{\mathbf{U}}_{i-1,j}}{\Delta x}, \frac{\bar{\mathbf{U}}_{i+1,j} - \bar{\mathbf{U}}_{i-1,j}}{2\Delta x}, \theta \frac{\bar{\mathbf{U}}_{i+1,j} - \bar{\mathbf{U}}_{i,j}}{\Delta x}\right), \\ (U_y)_{i,j} &= \text{minmod}\left(\theta \frac{\bar{\mathbf{U}}_{i,j} - \bar{\mathbf{U}}_{i,j-1}}{\Delta y}, \frac{\bar{\mathbf{U}}_{i,j+1} - \bar{\mathbf{U}}_{i,j-1}}{2\Delta y}, \theta \frac{\bar{\mathbf{U}}_{i,j+1} - \bar{\mathbf{U}}_{i,j}}{\Delta y}\right), \end{aligned} \quad (11)$$

where $\theta \in [1, 2]$ is a parameter used to determine the level of numerical dissipation. Note that larger values of θ correspond to less dissipation, leading to more oscillatory reconstructions.

The local speeds $a^\pm$ and $b^\pm$ are computed using the eigenvalues of the Jacobians $\frac{\partial \mathbf{F}}{\partial \mathbf{U}}$ and $\frac{\partial \mathbf{G}}{\partial \mathbf{U}}$, respectively, as follows:

$$\begin{aligned} a^+_{i+\frac{1}{2},j} &= \max\left\{u^-_{i+\frac{1}{2},j} + \sqrt{gh^-_{i+\frac{1}{2},j}}, u^+_{i+\frac{1}{2},j} + \sqrt{gh^+_{i+\frac{1}{2},j}}, 0\right\}, \\ a^-_{i+\frac{1}{2},j} &= \min\left\{u^-_{i+\frac{1}{2},j} - \sqrt{gh^-_{i+\frac{1}{2},j}}, u^+_{i+\frac{1}{2},j} - \sqrt{gh^+_{i+\frac{1}{2},j}}, 0\right\}, \end{aligned}$$

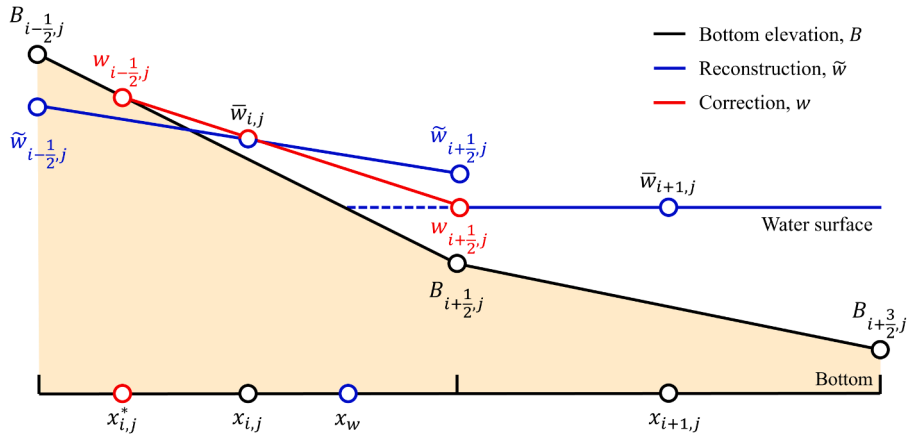


Fig. 1. Approximation of a piecewise linear reconstruction and correction for w at the wet-dry front.

$$b_{i,j+\frac{1}{2}}^+ = \max \left\{ v_{i,j+\frac{1}{2}}^- + \sqrt{gh_{i,j+\frac{1}{2}}^-}, v_{i,j+\frac{1}{2}}^+ + \sqrt{gh_{i,j+\frac{1}{2}}^+}, 0 \right\}, \quad (12)$$

$$b_{i,j+\frac{1}{2}}^- = \min \left\{ v_{i,j+\frac{1}{2}}^- - \sqrt{gh_{i,j+\frac{1}{2}}^-}, v_{i,j+\frac{1}{2}}^+ - \sqrt{gh_{i,j+\frac{1}{2}}^+}, 0 \right\}.$$

Kurganov and Levy [25] proposed a well-balanced upwind scheme with an appropriate discretization of the source term $\bar{S}_{i,j}(t)$, which is written as

$$\begin{aligned} \bar{S}_{i,j}^{(2)}(t) &= -g\bar{h}_{i,j} \frac{B(x_{i+\frac{1}{2}}, y_j) - B(x_{i-\frac{1}{2}}, y_j)}{\Delta x} \\ \bar{S}_{i,j}^{(3)}(t) &= -g\bar{h}_{i,j} \frac{B(x_i, y_{j+\frac{1}{2}}) - B(x_i, y_{j-\frac{1}{2}})}{\Delta y}. \end{aligned} \quad (13)$$

This semi-discrete scheme is applicable only when the bottom topography function B is continuous. Consequently, we replace the bottom topography function B by its continuous piecewise linear approximation $\tilde{B}$, defined as follows:

$$\begin{aligned} \tilde{B}(x, y) &= B_{i-\frac{1}{2}, j-\frac{1}{2}} + \left(B_{i+\frac{1}{2}, j-\frac{1}{2}} - B_{i-\frac{1}{2}, j-\frac{1}{2}} \right) \cdot \frac{x - x_{i-\frac{1}{2}}}{\Delta x} \\ &+ \left(B_{i-\frac{1}{2}, j+\frac{1}{2}} - B_{i-\frac{1}{2}, j-\frac{1}{2}} \right) \cdot \frac{y - y_{j-\frac{1}{2}}}{\Delta y} \\ &+ \left(B_{i+\frac{1}{2}, j+\frac{1}{2}} - B_{i+\frac{1}{2}, j-\frac{1}{2}} - B_{i-\frac{1}{2}, j+\frac{1}{2}} + B_{i-\frac{1}{2}, j-\frac{1}{2}} \right) \frac{(x - x_{i-\frac{1}{2}})(y - y_{j-\frac{1}{2}})}{\Delta x \Delta y}, \quad (x, y) \in I_{i,j}, \end{aligned} \quad (14)$$

where $B_{i\pm\frac{1}{2}, j\pm\frac{1}{2}}$ are the values of $\tilde{B}$ at the interfaces of cell $I_{i,j}$, which in case of a continuous function B at $(x_{i\pm\frac{1}{2}, j\pm\frac{1}{2}}, y_{i\pm\frac{1}{2}, j\pm\frac{1}{2}})$ reduces to $B_{i\pm\frac{1}{2}, j\pm\frac{1}{2}} = B(x_{i\pm\frac{1}{2}}, y_{j\pm\frac{1}{2}})$.

2.2. A two-dimensional reconstruction at partially flooded cells

A basic piecewise linear reconstruction may produce negative water depth values at the wet/dry fronts as the nonlinear flux limiter cannot ensure the positivity of the computed water depth, as shown in Fig. 1. At the wet/dry front, the reconstructed water surface elevation $\tilde{w}_{i-\frac{1}{2}, j}^+$ computed from the minmod limiter have a smaller value than the bottom elevation $B_{i-\frac{1}{2}, j}$. Consequently, the reconstructed (computed) water depth $\tilde{h}_{i-\frac{1}{2}, j}^+$ might be negative although the cell contains enough amount of water ($\tilde{w}_{i,j} > B_{i,j}$) as

$$\begin{aligned} \tilde{h}_{i-\frac{1}{2}, j}^+ &:= \tilde{w}_{i-\frac{1}{2}, j}^+ - B_{i-\frac{1}{2}, j} < 0, \\ \tilde{h}_{i+\frac{1}{2}, j}^- &:= \tilde{w}_{i+\frac{1}{2}, j}^- - B_{i-\frac{1}{2}, j} > 0. \end{aligned} \quad (15)$$

Therefore, it is imperative to correct the reconstructed water surface elevations to ensure that all computed water depths within the computational domain remain strictly non-negative. Bollermann et al. [16] proposed a one-dimensional correction method to

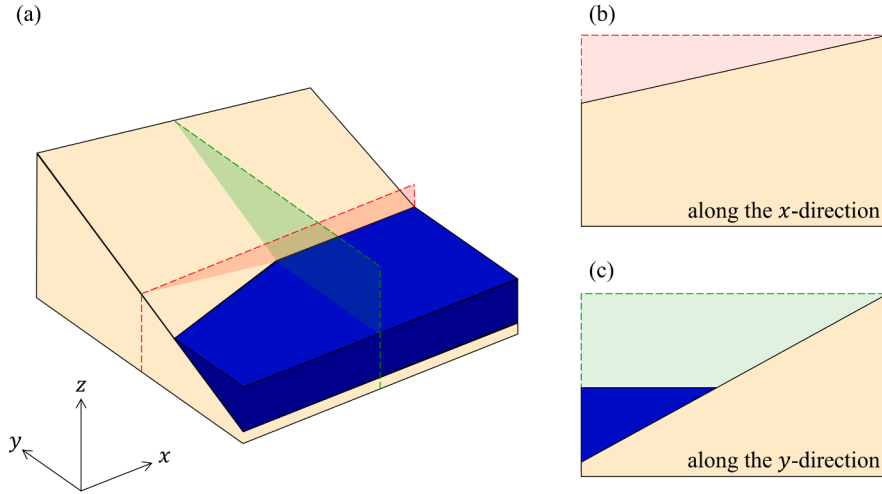


Fig. 2. 3D schematic diagram of partially flooded cells with $B_x \gg B_y$, including cross-sectional views in x and y directions.

guarantee positivity-preserving and well-balanced properties near the wet/dry boundary. We extend the scheme of Bollermann et al. to two dimensions, with particular attention to partially flooded cells exhibiting different bottom gradients B_x and B_y , along the x - and y -directions. The correction scheme exclusively adjusts the reconstructed water surfaces $\tilde{w}^\pm$, derived from the nonlinear flux limiter (piecewise linear reconstruction).

The situation within partially flooded cells becomes more intricate in two-dimensional space compared to one-dimensional space. For instance, when the bottom gradient along the x -direction is significantly steeper than that along the y -direction ($B_x \gg B_y$), the cell along the x -direction might be dry state despite containing a certain amount of water (see Fig. 2). Thus, to correctly reconstruct water surface elevations in two-dimensional spaces, it is essential to categorize cell states into 'dry' and 'non-dry' states. The 'non-dry' state can be further divided into fully and partially flooded cells. In this study, we introduce robust criteria for detecting dry cells using cell-averaged water depths $\bar{h}_{i,j}$, cell-averaged bottom elevations $B_{i,j}$, and bottom elevations at cell interfaces $B_{i\pm\frac{1}{2},j\pm\frac{1}{2}}$.

In one-dimensional space, these criteria can be categorized as follows: If $\bar{w}_i > B_i$, so $\bar{h}_i > 0$, then the cell I_i is non-dry. If $\bar{w}_i \leq B_i$, the cell I_i might be dry state, as shown in Fig. 2. In this case, if its neighboring cell I_{i+1} is fully flooded, i.e., $\bar{w}_{i+1} > B_{i+\frac{1}{2}}$ and $\bar{w}_{i+1} > B_{i+\frac{3}{2}}$, then the cell I_i is non-dry state. Otherwise, we set the cell I_i as dry state to prevent spurious errors resulting from truncation errors. Finally, if $\bar{w}_i = \min(B_{i-\frac{1}{2}}, B_{i+\frac{1}{2}})$, then the cell I_i is dry state. For dry cells, the piecewise linear reconstruction of water surface elevation is assumed to equal that of bottom elevation, thereby setting the reconstructed water depths at cell interfaces to zero. For non-dry cells at the wet/dry fronts, the following correction scheme is implemented.

The wet/dry correction scheme (refer to Bollermann et al. [16] for further details) can be briefly summarized using the following definitions: $w^\pm$ represent the corrected water surfaces, x_w^* denotes the intersection point between the water surface line and bathymetry, $x_{i,j}^*$ indicates the separation point between the wet and dry regions; i.e., $\Delta x_w^* = x_{i+\frac{1}{2},j} - x_w^*$ and $\Delta x_{i,j}^* = x_{i+\frac{1}{2},j} - x_{i,j}^*$, respectively.

If $\bar{w}_{i,j} > B_{i-\frac{1}{2},j}$ and $\bar{w}_{i,j} > B_{i+\frac{1}{2},j}$, then cell $I_{i,j}$ is fully flooded. If the reconstructed values of w at the cell interfaces are higher than B at the cell interfaces ($\tilde{w}_{i-\frac{1}{2},j}^+ \geq B_{i-\frac{1}{2},j}$ and $\tilde{w}_{i+\frac{1}{2},j}^- \geq B_{i+\frac{1}{2},j}$), we set

$$w_{i+\frac{1}{2},j}^\pm := \tilde{w}_{i+\frac{1}{2},j}^\pm. \quad (16)$$

Otherwise, we redistribute the surface water elevation. If $\tilde{w}_{i-\frac{1}{2},j}^+ < B_{i-\frac{1}{2},j}$, take $(w_x)_{i,j} := \frac{\tilde{w}_{i,j} - B_{i-\frac{1}{2},j}}{\Delta x/2}$, which leads to:

$$\begin{aligned} w_{i-\frac{1}{2},j}^+ &= B_{i-\frac{1}{2},j}, \\ w_{i+\frac{1}{2},j}^- &= 2\tilde{w}_{i,j} - B_{i-\frac{1}{2},j}. \end{aligned} \quad (17)$$

Likewise, if $\tilde{w}_{i+\frac{1}{2},j}^- < B_{i+\frac{1}{2},j}$, take $(w_x)_{i,j} := \frac{B_{i+\frac{1}{2},j} - \tilde{w}_{i,j}}{\Delta x/2}$, which leads to:

$$\begin{aligned} w_{i-\frac{1}{2},j}^+ &= 2\tilde{w}_{i,j} - B_{i+\frac{1}{2},j}, \\ w_{i+\frac{1}{2},j}^- &= B_{i+\frac{1}{2},j}. \end{aligned} \quad (18)$$

If $B_{i-\frac{1}{2},j} > \bar{w}_{i,j} > B_{i+\frac{1}{2},j}$, cell $I_{i,j}$ is possibly partially flooded. If its neighboring cell $C_{i+1,j}$ is fully flooded, i.e., $\bar{w}_{i+\frac{1}{2},j}^+ > B_{i+\frac{1}{2},j}$ and $\bar{w}_{i+\frac{3}{2},j}^- > B_{i+\frac{3}{2},j}$, then $w_{i+\frac{1}{2},j}^+ = \bar{w}_{i+\frac{1}{2},j}^+$. In this case, we set

$$\begin{aligned} w_{i+\frac{1}{2},j}^- &:= w_{i+\frac{1}{2},j}^+, \\ h_{i+\frac{1}{2},j}^- &:= w_{i+\frac{1}{2},j}^- - B_{i+\frac{1}{2},j}. \end{aligned} \quad (19)$$

If $2\bar{h}_{i,j} - h_{i+\frac{1}{2},j}^- \geq 0$, then the amount of water in cell $I_{i,j}$ is sufficient, and we set

$$\begin{aligned} h_{i-\frac{1}{2},j}^+ &= 2\bar{h}_{i,j} - h_{i+\frac{1}{2},j}^-, \\ w_{i-\frac{1}{2},j}^+ &= h_{i-\frac{1}{2},j}^+ + B_{i-\frac{1}{2},j}. \end{aligned} \quad (20)$$

Otherwise, we set

$$\begin{aligned} h_{i-\frac{1}{2},j}^+ &= 0, \\ w_{i-\frac{1}{2},j}^+ &= B_{i-\frac{1}{2},j}, \\ \Delta x_{i,j}^* &= \Delta x \cdot \min \left\{ \frac{2\bar{h}_{i,j}}{h_{i+\frac{1}{2},j}^-}, 1 \right\}. \end{aligned} \quad (21)$$

Meanwhile, if its neighboring cell is not fully flooded, we set

$$\begin{aligned} h_{i+\frac{1}{2},j}^- &:= \sqrt{2\bar{h}_{i,j} |B_{i-\frac{1}{2},j} - B_{i+\frac{1}{2},j}|}, \\ \Delta x_{i,j}^* &= \Delta x_w^* = \sqrt{\frac{2\Delta x \bar{h}_{i,j}}{-(B_x)_{i,j}}}. \end{aligned} \quad (22)$$

Note that this situation is not generic and may occur only in the under-resolved computations [16]. Finally, if $B_{i-\frac{1}{2},j} < \bar{w}_{i,j} < B_{i+\frac{1}{2},j}$, then this case is analogous to the possibly partially flooded cell, i.e., $B_{i-\frac{1}{2},j} > \bar{w}_{i,j} > B_{i+\frac{1}{2},j}$.

This reconstruction ensures the non-negativity of the reconstructed water depths, although these values may be zero or very small. To avoid cancellation issues, the flow velocities u and v are computed as:

$$u := \begin{cases} (\bar{h}u)/\bar{h}, & \text{if } \bar{h} \geq \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad v := \begin{cases} (\bar{h}v)/\bar{h}, & \text{if } \bar{h} \geq \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad (23)$$

where ϵ is a prescribed tolerance value. Using Eq. (23), the variables (hu) and (hv) can be decomposed into products of each variable:

$$(hu) := h \cdot u, \quad (hv) := h \cdot v. \quad (24)$$

2.3. Positivity-preserving property

In the previous section, we presented a two-dimensional reconstruction scheme for partially flooded cells. This section demonstrates the positivity-preserving property of the proposed scheme when the system (4) is discretized in time using either the forward Euler method or a higher-order strong stability preserving (SSP) ODE solver such as Runge-Kutta method. The first component of Eq. (4), along with the forward Euler time integration, can be rewritten as follows:

$$\bar{w}_{i,j}^{n+1} = \bar{w}_{i,j}^n - \lambda \left((H^x)_{i+\frac{1}{2},j}^{(1)} - (H^x)_{i-\frac{1}{2},j}^{(1)} \right) - \mu \left((H^y)_{i,j+\frac{1}{2}}^{(1)} - (H^y)_{i,j-\frac{1}{2}}^{(1)} \right), \quad (25)$$

where $\lambda := \Delta t / \Delta x$, $\mu := \Delta t / \Delta y$ and the superscript n represents the discretized time level, with the time level being $t^{n+1} = t^n + \Delta t$ for a time step Δt . Subtracting $B_{i,j}$ from Eq. (25) and substituting $\bar{w}_{i,j} - B_{i,j}$ with $\bar{h}_{i,j}$, one can obtain

$$\bar{h}_{i,j}^{n+1} = \bar{h}_{i,j}^n - \lambda \left((H^x)_{i+\frac{1}{2},j}^{(1)} - (H^x)_{i-\frac{1}{2},j}^{(1)} \right) - \mu \left((H^y)_{i,j+\frac{1}{2}}^{(1)} - (H^y)_{i,j-\frac{1}{2}}^{(1)} \right). \quad (26)$$

For the sake of clarity, the proof of the positivity-preserving property for fully flooded cells is omitted, as it has already been derived from Kurganov and Petrova [14]. It is concluded that the cell-averaged water depth at the time level $t = t^{n+1}$ is non-negative for all fully flooded cells under the following CFL condition:

$$\Delta t \leq \frac{1}{4} \min \left(\frac{\Delta x}{a}, \frac{\Delta y}{b} \right) \quad (27)$$

Table 1
 L^1 -errors and numerical orders of accuracy.

No. of cells	h		hu	
	L^1 -error	Order	L^1 -error	Order
100	5.18e-04	-	1.04e-02	-
200	1.46e-04	1.88	2.60e-03	2.00
400	3.66e-05	2.00	6.40e-04	2.02
800	9.07e-06	2.01	1.59e-04	2.00
1600	2.26e-06	2.00	3.97e-05	2.00
3200	5.57e-07	2.01	9.82e-06	2.01
6400	1.33e-07	2.05	2.34e-06	2.05

in which

$$a = \max_{i,j} \left\{ \max \left(a_{i+\frac{1}{2},j}^+, -a_{i+\frac{1}{2},j}^- \right) \right\}$$

$$b = \max_{i,j} \left\{ \max \left(b_{i,j+\frac{1}{2}}^+, -b_{i,j+\frac{1}{2}}^- \right) \right\} \quad (28)$$

The presence of partially flooded cells can lead to severe constraints on the CFL condition. Thus, the draining time step for two-dimensional space is introduced based on the draining time technique [e.g.15,16]. If the water amount associated with the cross-section passing through the cell center in the x-direction, defined as the product of the local water depth and the corresponding grid size, is greater than that in the y-direction, as shown in Fig. 3 (i.e., $\bar{h}_x \Delta x \geq \bar{h}_y \Delta y$),

$$\Delta t_{i,j}^{\text{drain},y} = \frac{\bar{h}_y \Delta y}{\max \left((H^x)^{(1)}_{i+\frac{1}{2},j}, 0 \right) + \max \left(-(H^x)^{(1)}_{i-\frac{1}{2},j}, 0 \right) + \max \left((H^y)^{(1)}_{i,j+\frac{1}{2}}, 0 \right) + \max \left(-(H^y)^{(1)}_{i,j-\frac{1}{2}}, 0 \right)},$$

$$\Delta t_{i,j}^{\text{drain},x} = \Delta t_{i,j}^{\text{drain},y} + \frac{\bar{h}_x \Delta x - \bar{h}_y \Delta y}{\max \left((H^x)^{(1)}_{i+\frac{1}{2},j}, 0 \right) + \max \left(-(H^x)^{(1)}_{i-\frac{1}{2},j}, 0 \right)} \quad (29)$$

where the draining time step $\Delta t_{i,j}^{\text{drain}}$ represents the duration required for the cell $I_{i,j}$ to become dry through the outgoing fluxes. Consequently, the time evolution of water depth in Eq. (26) can be rewritten as

$$\bar{h}_{i,j}^{n+1} = \bar{h}_{i,j}^n - \frac{\left((H^x)^{(1)}_{i+\frac{1}{2},j} \Delta t_{i+\frac{1}{2},j} - (H^x)^{(1)}_{i-\frac{1}{2},j} \Delta t_{i-\frac{1}{2},j} \right)}{\Delta x}$$

$$- \frac{\left((H^y)^{(1)}_{i,j+\frac{1}{2}} \Delta t_{i,j+\frac{1}{2}} - (H^y)^{(1)}_{i,j-\frac{1}{2}} \Delta t_{i,j-\frac{1}{2}} \right)}{\Delta y}, \quad (30)$$

in which the effective time step $\Delta t_{i+\frac{1}{2},j}$ at the interface between cell $I_{i,j}$ and cell $I_{i+1,j}$ is determined by

$$\Delta t_{i+\frac{1}{2},j} = \begin{cases} \min(\Delta t, \Delta t_{i,j}^{\text{drain}}), & \text{if } \nabla \cdot H_{i,j}^{(1),x} > 0, \\ \min(\Delta t, \Delta t_{i+1,j}^{\text{drain}}), & \text{if } \nabla \cdot H_{i,j}^{(1),x} \leq 0, \end{cases} \quad (31)$$

which indicates that the effective time step is applied when outgoing fluxes dominate, potentially leading cell $I_{i,j}$ to dry out. If the flux divergence ($\nabla \cdot H_{i,j}^{(1)}$) is positive, implying the cell could become dry, the effective time step is determined as the lesser of the global time step and the draining time step. For partially flooded cells, the draining time step is shorter than the global time step, so it is utilized exclusively for the partially flooded cell when $\nabla \cdot H_{i,j}^{(1)} > 0$. As the draining time step locally suppresses numerical fluxes only to prevent negative water depths, without introducing artificial sources or sinks, the finite-volume formulation remains strictly conservative. Consequently, mass conservation is preserved even in partially flooded cells, while the positivity-preserving property is simultaneously ensured. The proof is still valid if the forward Euler method is replaced by a higher-order SSP ODE solvers, because such time discretizations can be decomposed into convex combinations of several forward Euler steps [26].

2.4. Well-balanced property

This section demonstrates the proof that the proposed scheme can guarantee the well-balanced property for the SWEs. To ensure well-balancing, the gravity-driven component of the momentum flux must balance with the source term in the stationary steady state,

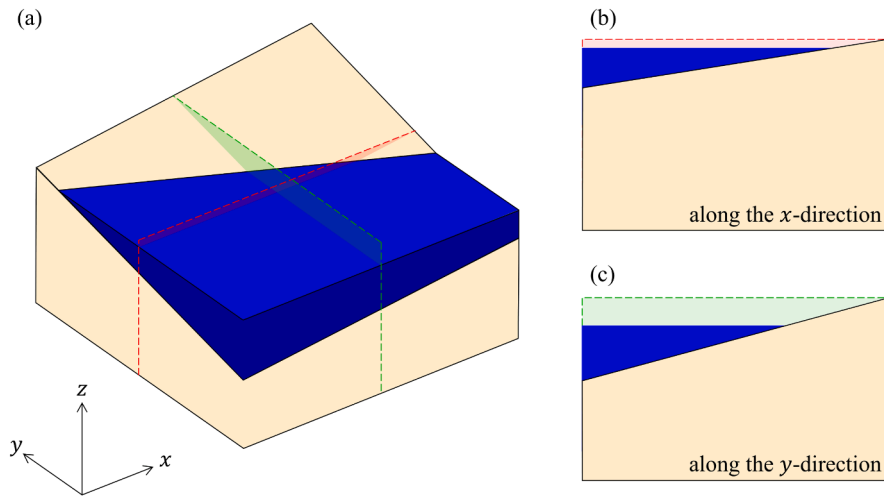


Fig. 3. 3D schematic diagram of partially flooded cells, including cross-sectional views in x and y directions.

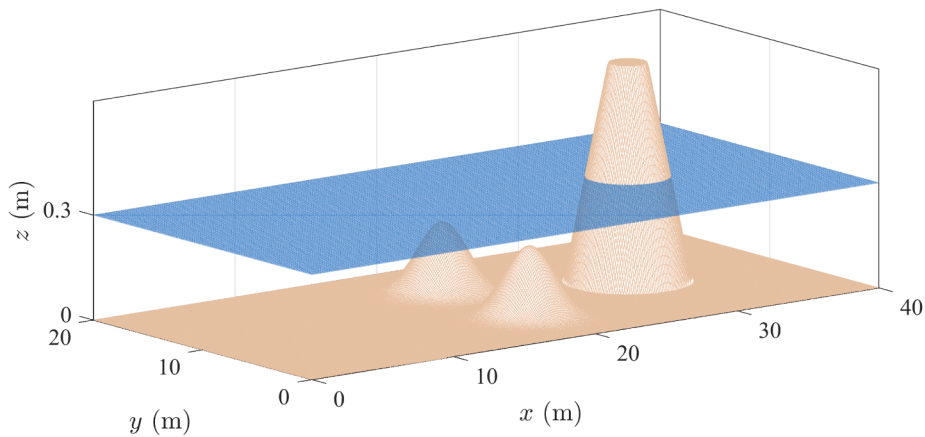


Fig. 4. Example 2: Initial water surface level with the bottom topography.

i.e., the “lake at rest”. Starting from Eq. (26), the second and third components of Eq. (4) for the “lake at rest” cases can be written as

$$\begin{aligned}\bar{h}u_{i,j}^{n+1} &= \bar{h}u_{i,j}^n - \lambda \left(\left(\frac{1}{2}gh^2 \right)_{i+\frac{1}{2},j} - \left(\frac{1}{2}gh^2 \right)_{i-\frac{1}{2},j} \right) \\ &\quad - \lambda g \left(\bar{w}_{i,j} - B_{i,j} \right) \left(B_{i+\frac{1}{2},j} - B_{i-\frac{1}{2},j} \right), \\ \bar{h}v_{i,j}^{n+1} &= \bar{h}v_{i,j}^n - \mu \left(\left(\frac{1}{2}gh^2 \right)_{i,j+\frac{1}{2}} - \left(\frac{1}{2}gh^2 \right)_{i,j-\frac{1}{2}} \right) \\ &\quad - \mu g \left(\bar{w}_{i,j} - B_{i,j} \right) \left(B_{i,j+\frac{1}{2}} - B_{i,j-\frac{1}{2}} \right).\end{aligned}\tag{32}$$

The above equations can be rewritten as

$$\begin{aligned}\bar{h}u_{i,j}^{n+1} &= \bar{h}u_{i,j}^n - \lambda \left(\frac{1}{2}g \left(h_{i+\frac{1}{2},j} \right)^2 - \frac{1}{2}g \left(h_{i-\frac{1}{2},j} \right)^2 \right) - \lambda g \bar{h}_{i,j} \left(B_{i+\frac{1}{2},j} - B_{i-\frac{1}{2},j} \right), \\ \bar{h}v_{i,j}^{n+1} &= \bar{h}v_{i,j}^n - \mu \left(\frac{1}{2}g \left(h_{i,j+\frac{1}{2}} \right)^2 - \frac{1}{2}g \left(h_{i,j-\frac{1}{2}} \right)^2 \right) - \mu g \bar{h}_{i,j} \left(B_{i,j+\frac{1}{2}} - B_{i,j-\frac{1}{2}} \right).\end{aligned}\tag{33}$$

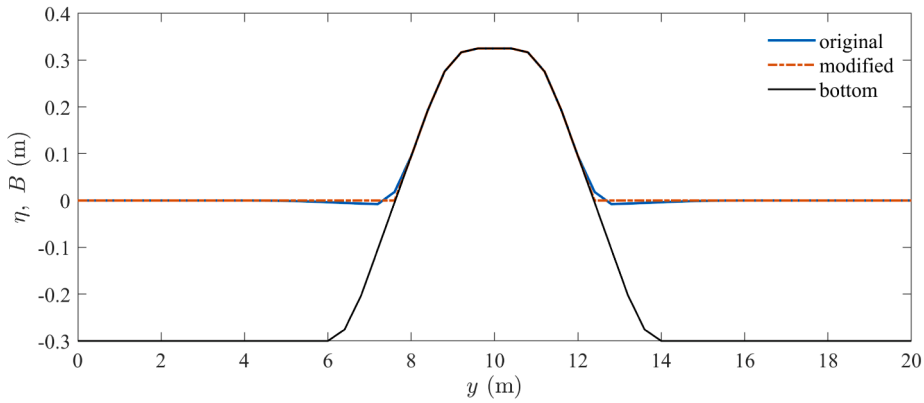


Fig. 5. Example 2: Water surface elevation profile along the cross-section $x = 30$ m at $t = 50$ s in “lake at rest” state.

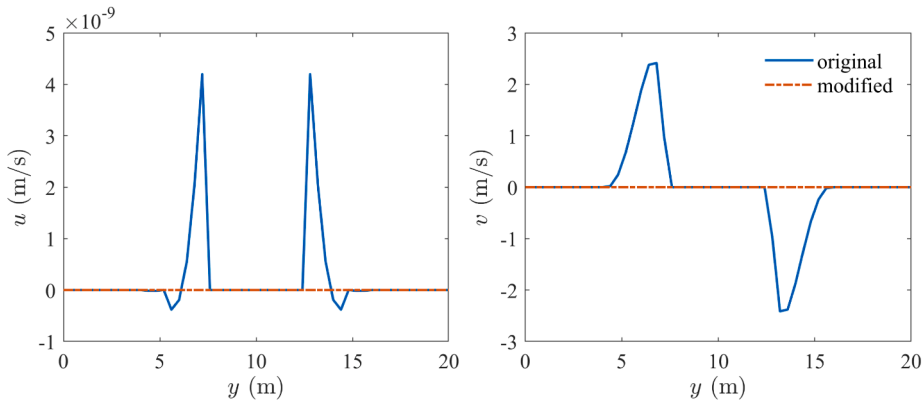


Fig. 6. Example 2: Velocity profiles along the cross-section $x = 30$ m at $t = 50$ s in “lake at rest” state.

In the fully flooded cells, we have

$$\begin{aligned}
 \frac{\left(h_{i+\frac{1}{2},j}\right)^2 - \left(h_{i-\frac{1}{2},j}\right)^2}{2} &= \frac{h_{i+\frac{1}{2},j} + h_{i-\frac{1}{2},j}}{2} \left(h_{i+\frac{1}{2},j} - h_{i-\frac{1}{2},j}\right) \\
 &= \bar{h}_{i,j} \left(w_{i,j} - B_{i+\frac{1}{2},j} - w_{i,j} + B_{i-\frac{1}{2},j}\right) \\
 &= -\bar{h}_{i,j} \left(B_{i+\frac{1}{2},j} - B_{i-\frac{1}{2},j}\right), \\
 \frac{\left(h_{i,j+\frac{1}{2}}\right)^2 - \left(h_{i,j-\frac{1}{2}}\right)^2}{2} &= \frac{h_{i,j+\frac{1}{2}} + h_{i,j-\frac{1}{2}}}{2} \left(h_{i,j+\frac{1}{2}} - h_{i,j-\frac{1}{2}}\right) \\
 &= \bar{h}_{i,j} \left(w_{i,j} - B_{i,j+\frac{1}{2}} - w_{i,j} + B_{i,j-\frac{1}{2}}\right) \\
 &= -\bar{h}_{i,j} \left(B_{i,j+\frac{1}{2}} - B_{i,j-\frac{1}{2}}\right).
 \end{aligned} \tag{34}$$

In the partially flooded cells, where $h_{i-\frac{1}{2},j} = 0$ or $h_{i,j-\frac{1}{2}} = 0$, Eq. (34) reduces to

$$\begin{aligned}
 \frac{\left(h_{i+\frac{1}{2},j}\right)^2}{2} &= -\frac{\Delta x_{i,j}^* h_{i+\frac{1}{2},j}}{2\Delta x} \left(B_{i+\frac{1}{2},j} - B_{i-\frac{1}{2},j}\right) = -\frac{h_{i+\frac{1}{2},j}}{2} \Delta x_{i,j}^* (B_x)_{i,j}, \\
 \frac{\left(h_{i,j+\frac{1}{2}}\right)^2}{2} &= -\frac{\Delta y_{i,j}^* h_{i,j+\frac{1}{2}}}{2\Delta y} \left(B_{i,j+\frac{1}{2}} - B_{i,j-\frac{1}{2}}\right) = -\frac{h_{i,j+\frac{1}{2}}}{2} \Delta y_{i,j}^* (B_y)_{i,j}.
 \end{aligned} \tag{35}$$

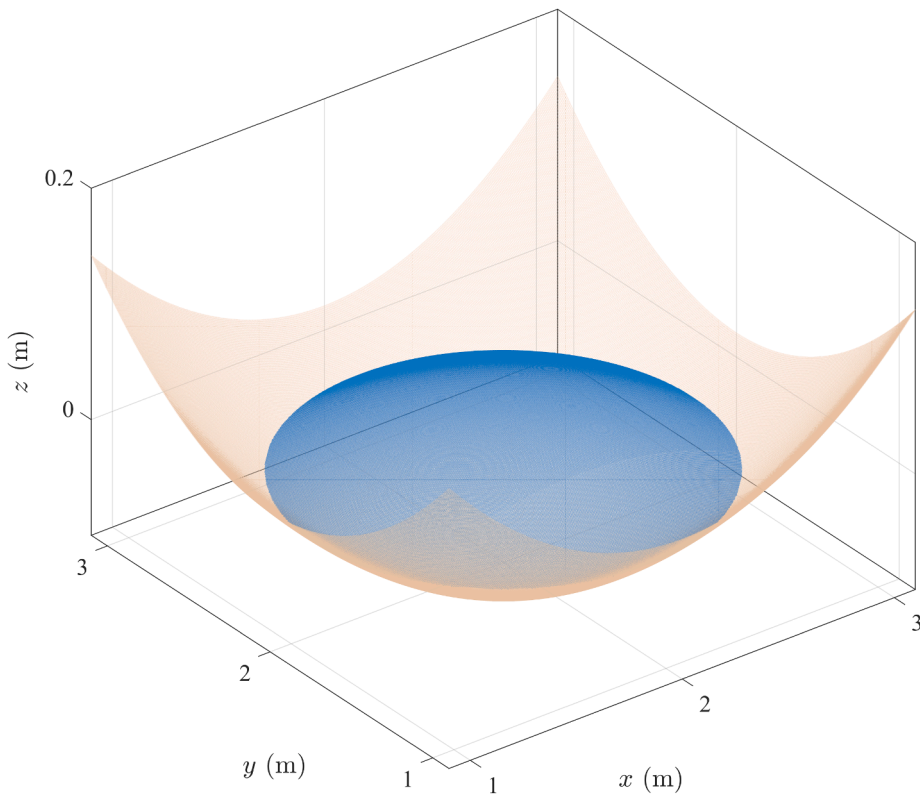


Fig. 7. Example 3.1: Initial water surface level with the bottom topography.

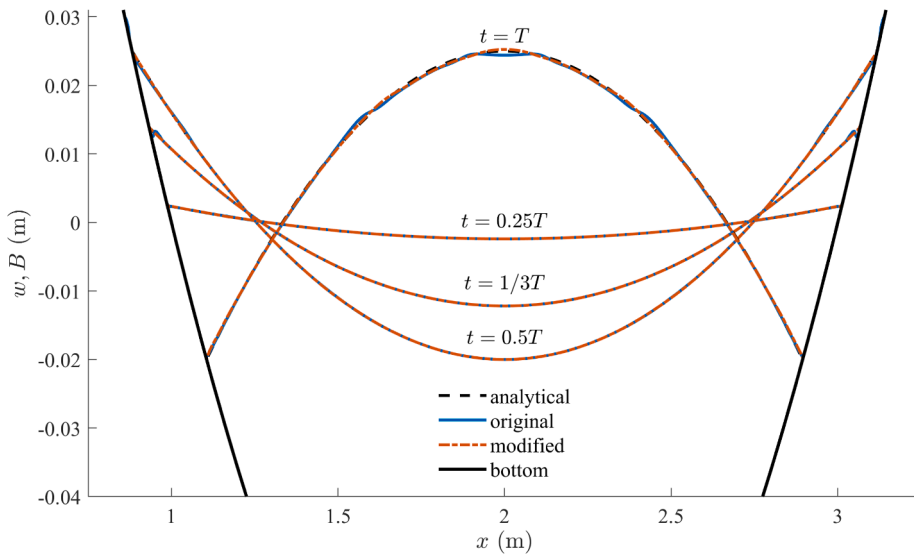


Fig. 8. Example 3.1: Water surface elevation profiles at different times.

The gravity-driven components of the momentum flux cancel the source terms in partially flooded cells as well as fully flooded cells. Therefore, Eq. (32) for the “lake at rest” cases reduces to

$$\begin{aligned} (\bar{h}u)_{i,j}^{n+1} &= (\bar{h}u)_{i,j}^n, \\ (\bar{h}v)_{i,j}^{n+1} &= (\bar{h}v)_{i,j}^n, \end{aligned} \quad (36)$$

which proves the well-balanced property of the proposed scheme for the SWEs.

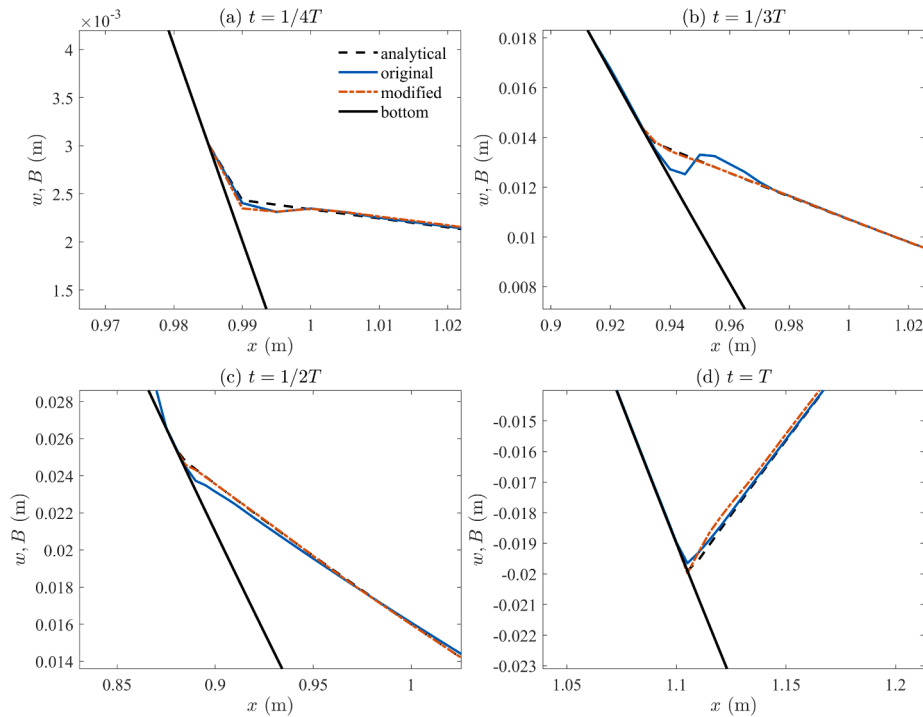


Fig. 9. Example 3.1: Water surface elevation profiles at different times, zoom at the wet/dry fronts.

3. Numerical experiments

In this section, we assess the performance of the proposed scheme against several numerical experiments with analytical and reference solutions. Moreover, laboratory experiments of solitary wave propagation in one- or two-dimensional spaces were numerically simulated. In all numerical experiments, the minmod parameter θ in Eq. (11) is set to 2.0, the gravitational acceleration g is 9.81 m/s^2 , and ϵ is set to 10^{-12} , except in Example 3.2, where ϵ is set to 10^{-16} .

3.1. Example 1—Numerical order of accuracy

In the first example taken from Xing and Shu [27], we verify the numerical order of accuracy of the proposed scheme for a continuous solution. This example is subject to the following initial conditions and bottom topography:

$$\begin{aligned} B(x, 0) &= \sin^2(\pi x), \quad h(x, 0) = 5 + \exp\{\cos(2\pi x)\}, \\ hu(x, 0) &= \sin(\cos(2\pi x)), \quad x \in [0, 1] \end{aligned} \quad (37)$$

with the periodic boundary conditions.

Although the exact solution to this example is not available, it serves as a generic test case for numerical accuracy since some known exact solutions (in closed form) have special properties, making the leading terms in the truncation errors in many schemes vanish [e.g. 27,28]. Consequently, this example has been widely used for model validation [e.g. 14,16,19,29,30]. The reference solution is computed using the same numerical scheme with 25,600 cells and is utilized as the exact solution to calculate numerical errors and numerical orders of accuracy for h and hu . It is computed up to $t = 0.1 \text{ s}$ when the solution is still smooth [27]. Table 1 presents the L^1 -errors and numerical orders of accuracy for this example, confirming that the proposed scheme has second-order accuracy.

3.2. Example 2—“Lake at rest” steady-state with wet-dry fronts

The purpose of this example, taken from [31], is to demonstrate the well-balanced property of the proposed scheme in the vicinity of wet/dry fronts. We take the computational domain with a length of 40 m in the x direction and a width of 20 m in the y direction, the bottom topography:

$$B(x, y) = \max\{B1(x, y), B2(x, y), B3(x, y), 0\}, \quad (38)$$

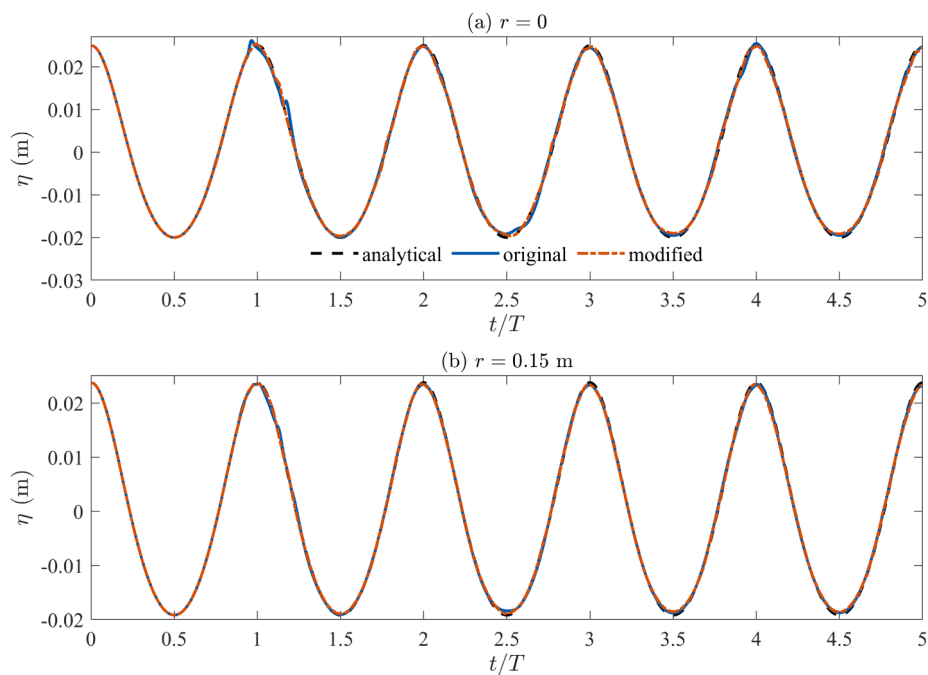


Fig. 10. Example 3.1: Time history of Water surface elevation at two different locations: (a) $r = 0$ and (b) $r = 0.15$ m.

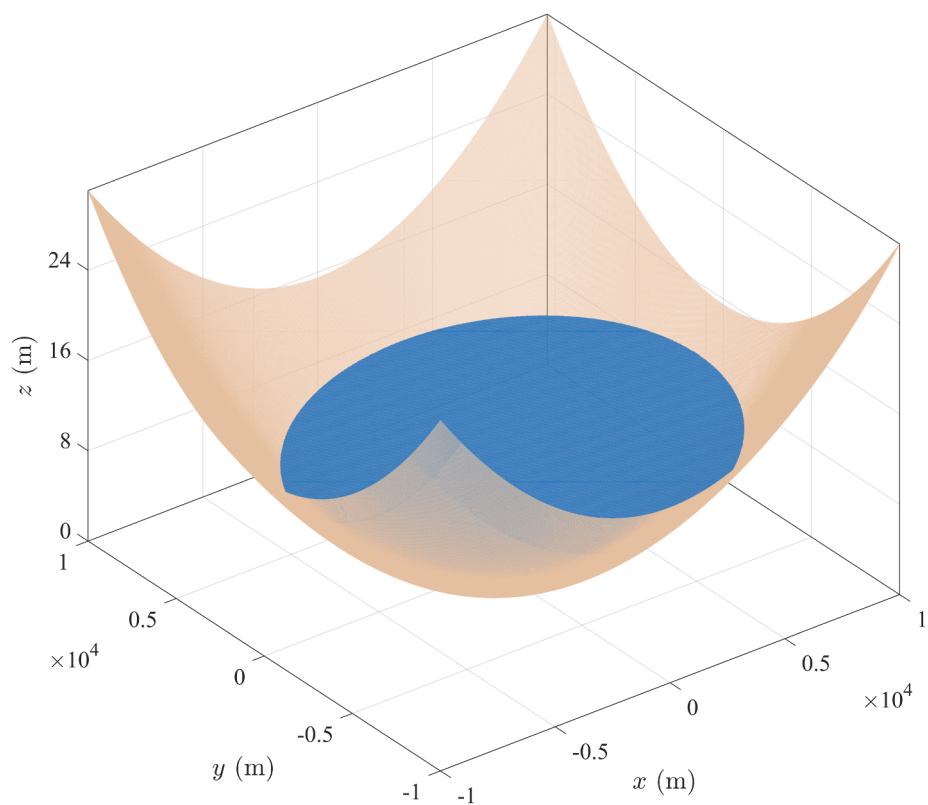


Fig. 11. Example 3.2: Initial water surface level with the bottom topography.

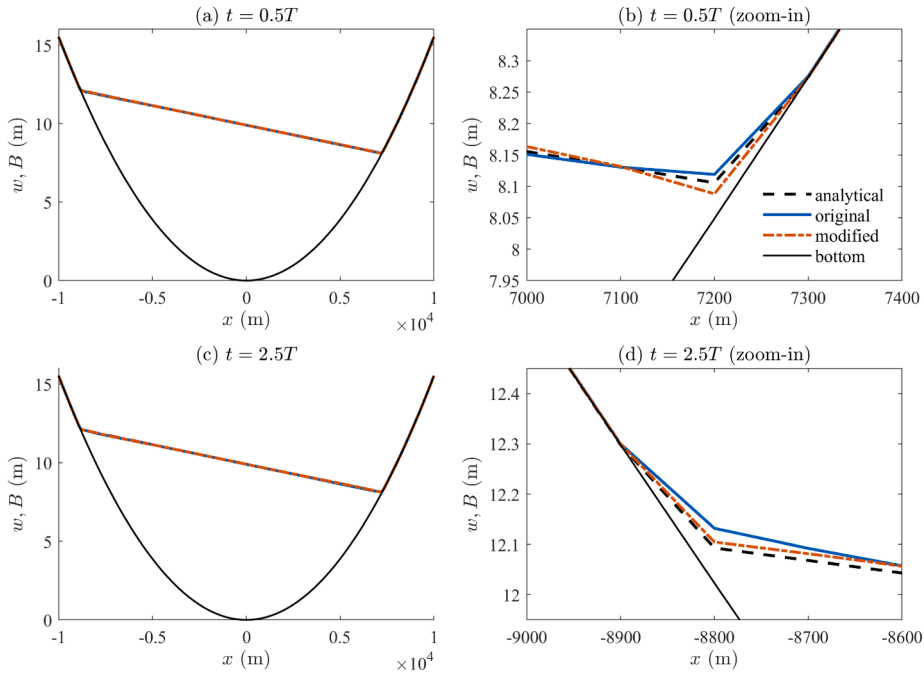


Fig. 12. Example 3.2: Water surface elevation profiles at $t = 0.5T$ and $t = 2.5T$, with enlarged views near the wet/dry boundaries.

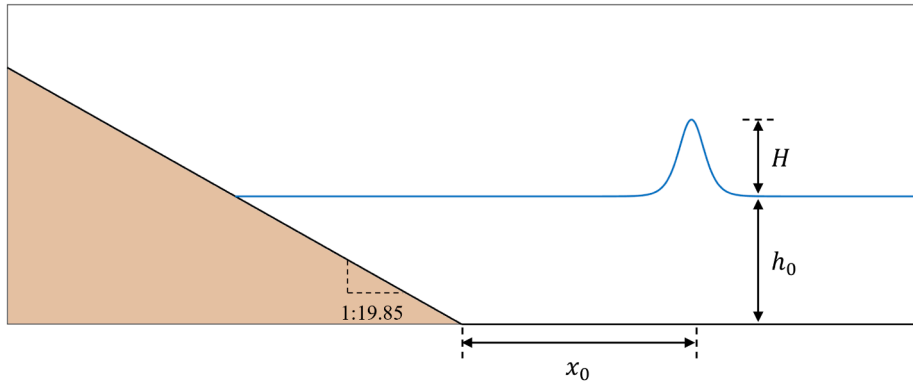


Fig. 13. Example 4: 1D schematic illustration of a solitary wave over a mild slope. The plot is not to scale.

where a conical island is defined by

$$B1(x, y) = \begin{cases} 0.625, & r \leq 1.1, \\ 0.25(3.6 - r), & 1.1 < r \leq 3.6, \quad r := \sqrt{(x - 30)^2 + (y - 10)^2} \\ 0, & \text{otherwise,} \end{cases} \quad (39)$$

and two mild humps are defined by

$$\begin{aligned} B2(x, y) &= 0.2 \exp \left\{ -0.25(x - 20)^2 - 0.25(y - 6)^2 \right\}, \\ B3(x, y) &= 0.2 \exp \left\{ -0.25(x - 20)^2 - 0.25(y - 14)^2 \right\}. \end{aligned} \quad (40)$$

the following initial conditions:

$$w(x, y, 0) = \max\{0.3, B(x, y)\}, \quad u(x, y, 0) = v(x, y, 0) = 0. \quad (41)$$

and the solid wall boundaries. A grid size of 0.4 m is used and we compute the solution until the time $t = 50$ s. The initial water surface level with the bottom topography is shown in Fig. 4.

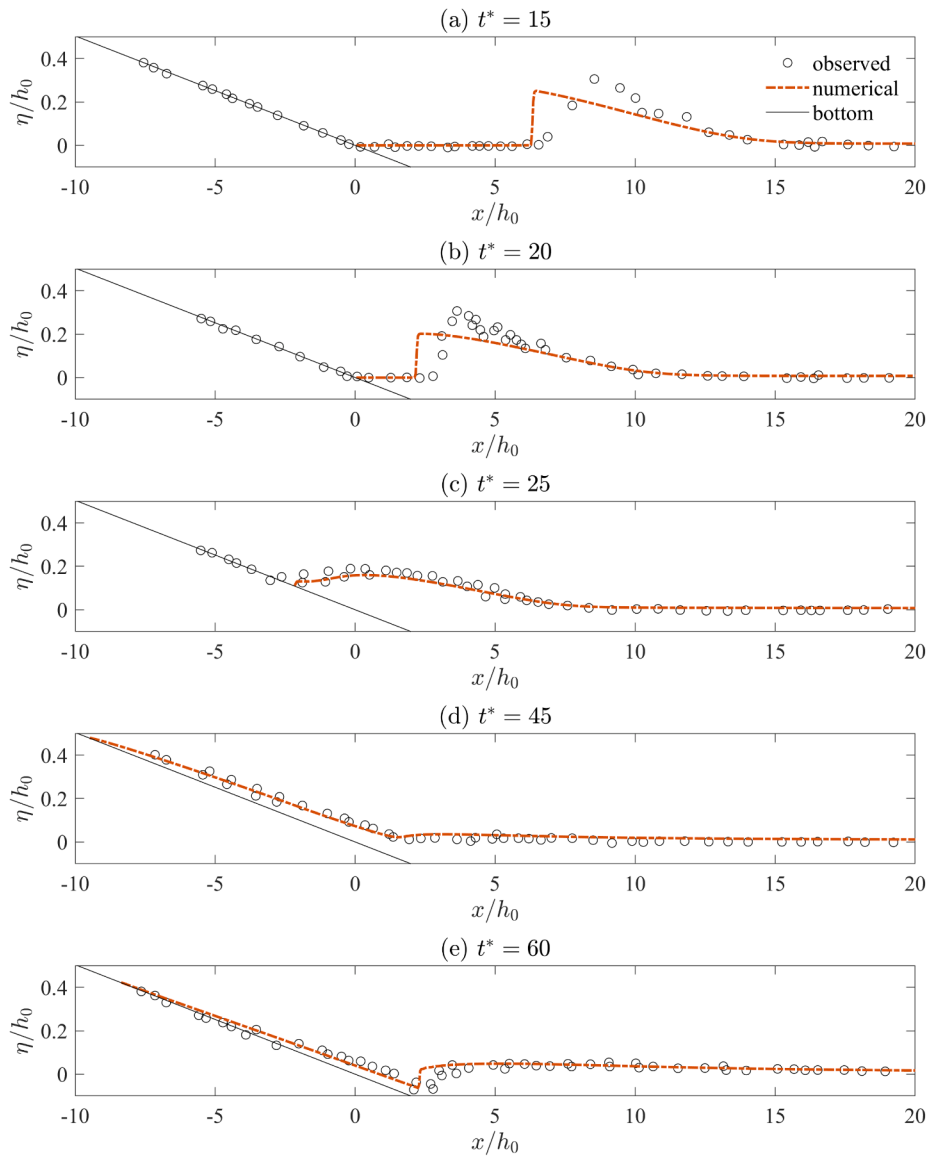


Fig. 14. Example 4: Temporal evolution of breaking solitary wave with $H/h_0 = 0.28$ over a mild slope.

Fig. 5 illustrates the water surface elevation profiles along the cross-section $x = 30$ m at $t = 50$, as computed by both numerical schemes. The original scheme fails to maintain the stationary steady state in the presence of wet/dry fronts, showing an error of $O(10^{-2})$. In contrast, the proposed scheme successfully preserves the initial state without significant oscillations, exhibiting an error of $O(10^{-17})$, which corresponds to the round-off error. Additionally, Fig. 6 shows the computed flow velocities u and v along the cross-section $x = 30$ m. It reveals that the original scheme produces spurious oscillations at the wet/dry interfaces, whereas the proposed scheme maintains the stationary steady state without artificial fluxes at wet/dry fronts. This demonstrates the ability of the proposed scheme to deal with wet/dry problems, in a non-oscillatory manner.

3.3. Example 3—Thacker's solutions

Among the various analytical solutions for nonlinear and horizontally two-dimensional problems, Thacker [32] presented several for time-dependent periodic oscillations in parabolic basins. These solutions involve periodic wetting and drying processes over the entire domain and thus are a complex test for validating the performance of numerical schemes against wet/dry fronts.

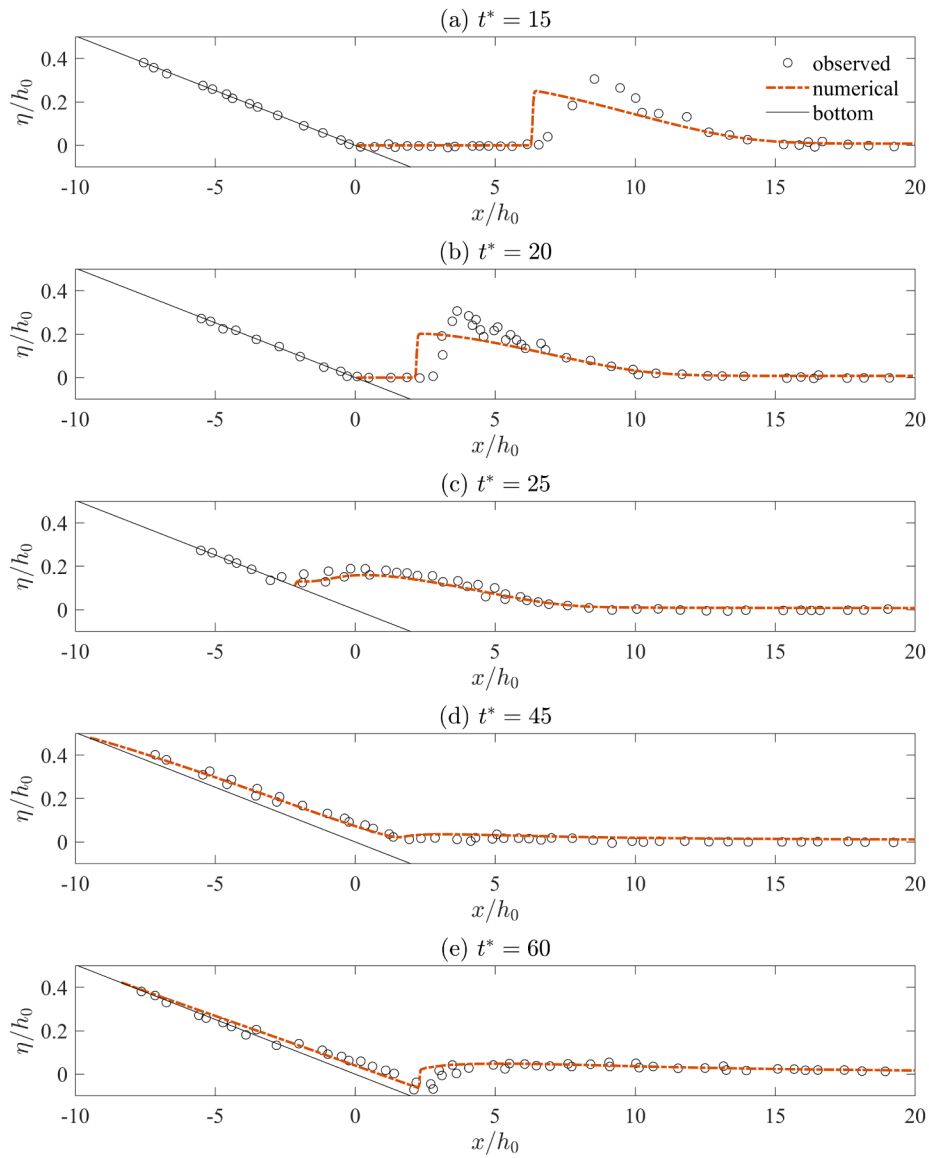


Fig. 15. Example 4: Temporal evolution of breaking solitary wave with $H/h_0 = 0.30$ over a mild slope.

3.3.1. Example 3.1—Long wave resonance in a parabolic basin

One of these solutions is a long wave resonance in a parabolic basin where the bottom topography is defined by

$$B(x, y) = -h_0 \left(1 - \frac{r^2}{a^2} \right), \quad (42)$$

and the analytical solutions of the water surface elevations are given by

$$w(x, y, t) = h_0 \left[\frac{(1 - A^2)^{1/2}}{1 - A \cos \omega t} - 1 - \frac{r^2}{a^2} \left\{ \frac{1 - A^2}{(1 - A \cos \omega t)^2} - 1 \right\} \right], \quad (43)$$

where

$$A = \frac{a^4 - r_0^4}{a^4 + r_0^4}, \quad \omega = \frac{1}{a} (8gh_0)^{0.5}, \quad (44)$$

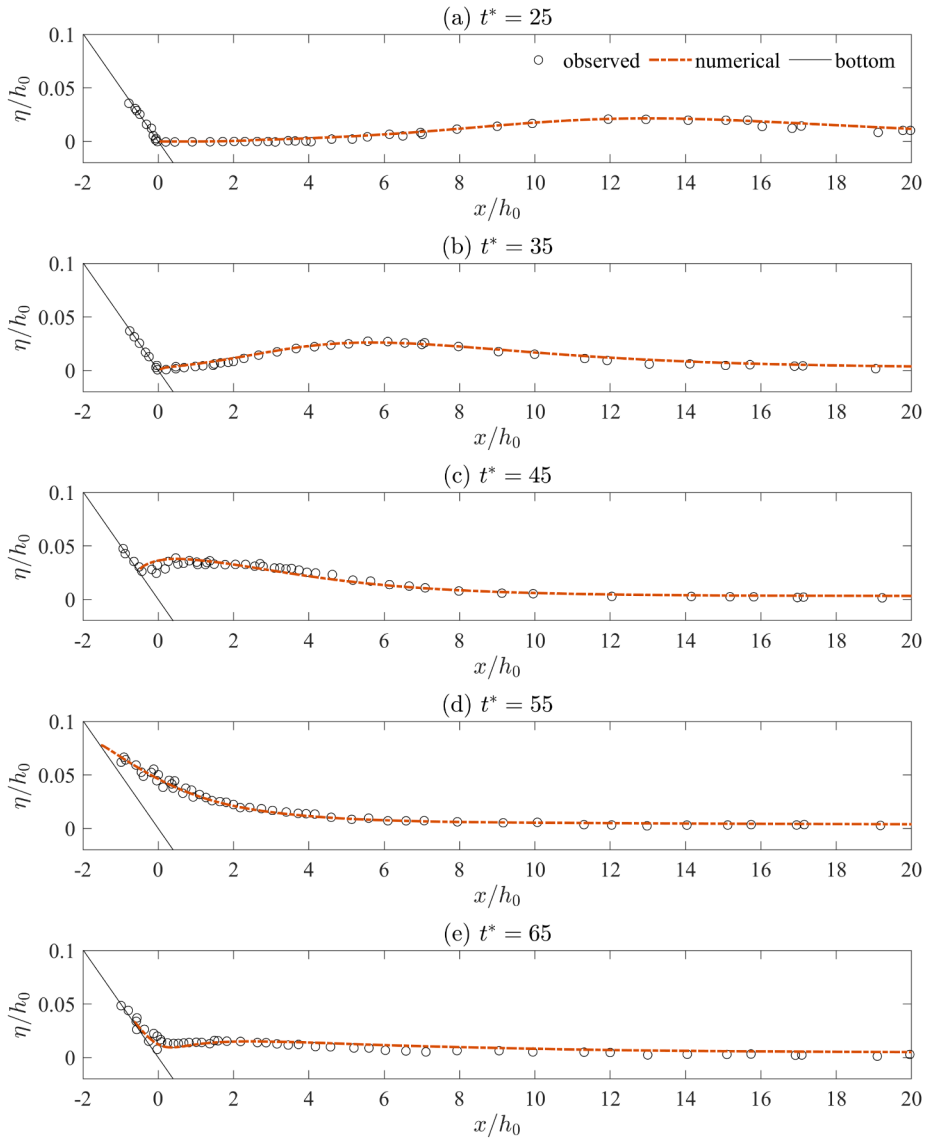


Fig. 16. Example 4: Temporal evolution of non-breaking solitary wave with $H/h_0 = 0.0185$ over a mild slope.

in which r is the distance from the center, h_0 is the initial water depth at the center ($r = 0$), a is the distance from the center to the shoreline with zero elevation, and r_0 is the distance from the center to the initial shoreline where the initial water depth is set to zero. In the numerical test, $h_0 = 0.1$ m, $a = 1$ m, and $r_0 = 0.8$ m. The computational domain is discretized using a grid size of 0.005 m. The initial water surface level with the bottom topography is shown in Fig. 7.

Fig. 8 depicts water surface elevation profiles at different times. Both numerical schemes show good agreement with the analytical solutions, though the original scheme exhibits a slight discrepancy. At $t = T$, the original scheme yields a relative error of 2.95% at $r = 0$, whereas the proposed scheme has an error of 1.24%. The relative error is computed as $|w_{\text{num}}(x, y, t) - w_{\text{ana}}(x, y, t)|/|w_{\text{ana}}(x, y, t)|$ and is evaluated here at $(x, y, t) = (2, 2, T)$, corresponding to $r = 0$, where w_{num} and w_{ana} denote the numerical and analytical solutions of w , respectively. However, at $t = 0.5T$, both schemes yield similar errors of less than 0.003% at $r = 0$. Fig. 9 further illustrates the enlarged views of water surface elevation profiles near the wet/dry fronts, highlighting the performance of the numerical schemes in solving a moving boundary problem. Notably, Fig. 9 (b) reveals spurious oscillations near the wet/dry fronts even during the first oscillation period when applying the original scheme.

Fig. 10 illustrates the temporal changes in water surface elevation over five periods at two distinct locations: the center ($r = 0$) and a point 0.15 m from the center ($r = 0.15$ m). The results indicate that the original scheme induces the undesired numerical fluctuations, especially at $t = T$ whereas the proposed scheme offers a better approximation without significant numerical oscillations. However,

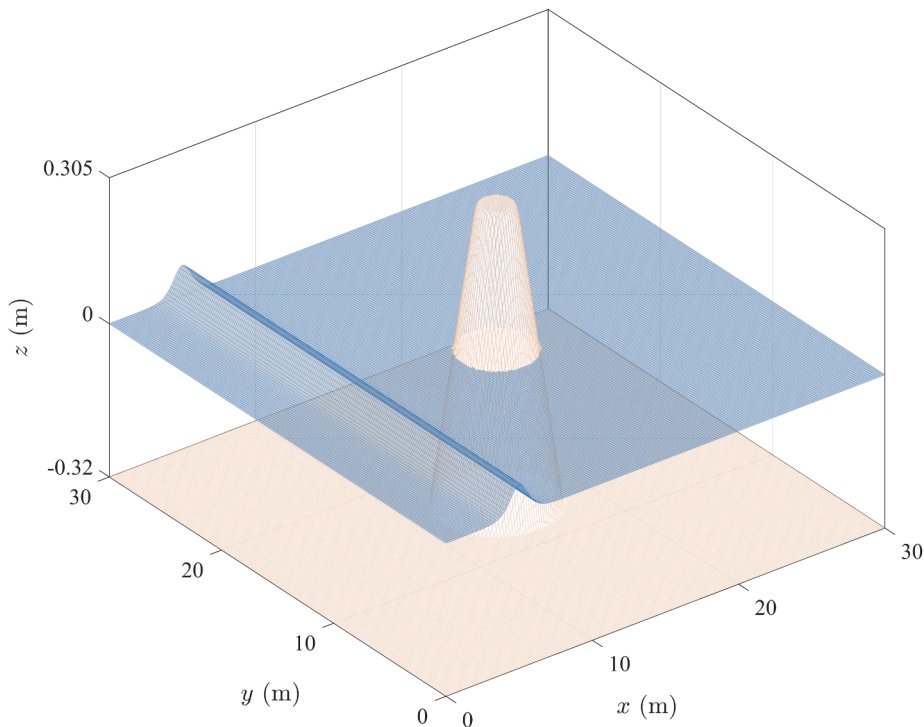


Fig. 17. Example 5: Initial solitary wave profile over varying bottom topography.

a numerical error, which is difficult to eliminate, gradually accumulates near the wet/dry interfaces over time. Similar results have been commonly observed in previous studies [e.g.14,17,33].

3.3.2. Example 3.2—Oscillating lake in a parabolic basin

Another solution, taken from [32], is a planar surface rotating around a parabolic basin. The bottom topography is defined by

$$B(x, y) = h_0 \left(\frac{x^2 + y^2}{a^2} \right), \quad (45)$$

and the analytical solutions are given by

$$\begin{aligned} w(x, y, t) &= h_0 \left[1 + \frac{\psi}{a^2} (2x \cos \omega t + 2y \sin \omega t - \psi) \right], \\ u(x, y, t) &= -\psi \omega \sin \omega t, \\ v(x, y, t) &= -\psi \omega \cos \omega t \end{aligned} \quad (46)$$

where $h_0 = 10$ m, $a = 8025.5$ m, $\psi = a/10$, and the period $T = 3600$ s. A uniform grid size of 100 m is adopted in x and y directions to discretize the computational domain. Fig. 11 displays the initial water surface level along with the bottom topography.

Fig. 12 compares the water surface elevation profiles at $t = 0.5T$ and $t = 2.5T$, including enlarged views near the wet/dry boundaries. As in Example 3.1, we can see that the numerical solutions from both schemes converge well with the analytical solutions across the entire numerical domain, without notable numerical fluctuations near the wet/dry boundaries. Furthermore, Fig. 12 (d) suggests that the modified scheme aligns more closely with the analytical solution, particularly by reducing the accumulating numerical oscillations near the wet/dry boundaries during multiple wetting and drying cycles. This indicates improved stability and accuracy near the wet/dry fronts. These two parabolic basin examples confirm the accuracy and stability of the proposed numerical scheme in handling periodic wetting and drying processes in two-dimensional space.

3.4. Example 4—One-dimensional solitary wave run-up on a mild slope

This example, proposed by Synolakis [34], is a run-up and reflection of a solitary wave on a mild slope. The experiment domain involves an inclined frictional plane with a slope of 1 : 19.85, which transitions into a region of constant depth. A solitary wave of height H is generated at $x = x_0$ from the toe of the slope. The initial conditions are given by

$$\eta(x) = H \operatorname{sech}^2 \left[\sqrt{\frac{3H}{4h_0^3}} (x - x_0) \right],$$

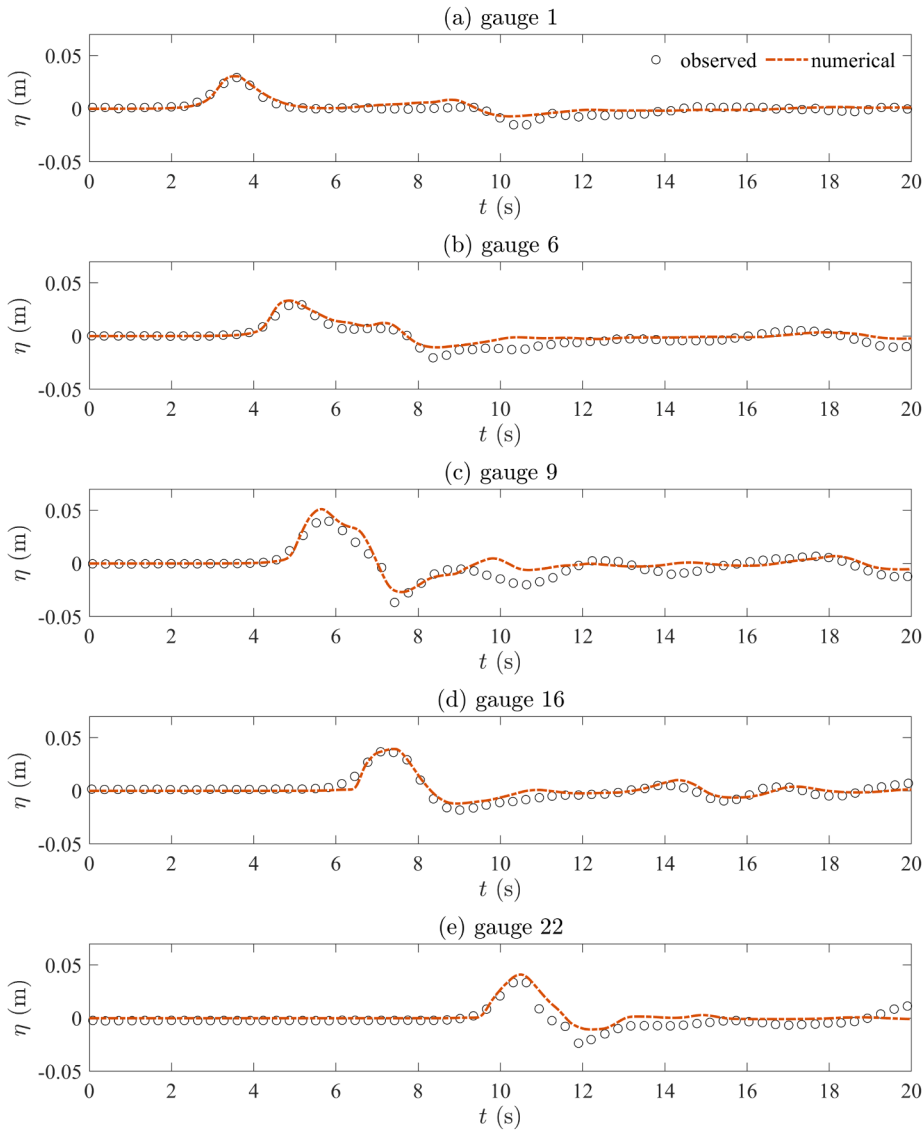


Fig. 18. Example 5: Temporal evolution of solitary wave with $H/h_0 = 0.1$ over the conical island.

$$u(x) = \sqrt{\frac{3H}{4h_0^3}} \left(\frac{\eta}{\eta + h_0} \right). \quad (47)$$

where h_0 is the water depth at a constant depth region. The distance x_0 from the toe of the slope is given by [34]:

$$x_0 = \sqrt{\frac{4h_0}{3H}} \operatorname{arccosh} \left[\left(\frac{1}{0.05} \right)^{\frac{1}{2}} \right]. \quad (48)$$

In this example, the Manning formula is utilized to account for the friction effect; correspondingly, the governing Eq. (4) are modified as follows:

$$\begin{cases} w_t + (hu)_x + (hv)_y = 0, \\ (hu)_t + (hu^2 + \frac{1}{2}gh^2)_x + (huv)_y = -ghB_x - \frac{gn^2u\sqrt{u^2+v^2}}{h^{1/3}}, \\ (hv)_t + (huv)_x + (hv^2 + \frac{1}{2}gh^2)_y = -ghB_y - \frac{gn^2v\sqrt{u^2+v^2}}{h^{1/3}}, \end{cases} \quad (49)$$

where n is the Manning roughness coefficient. In this example, $n = 0.01$ is used.

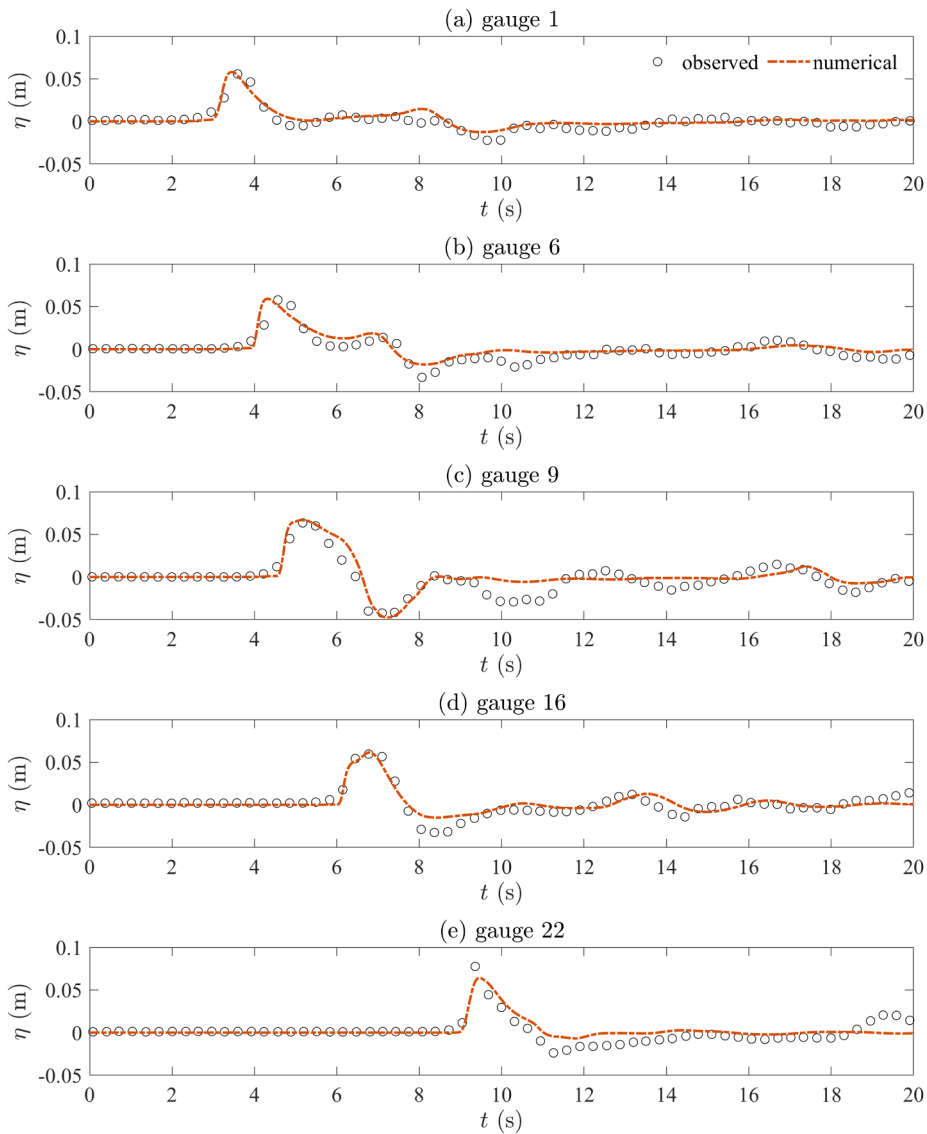


Fig. 19. Example 5: Temporal evolution of solitary wave with $H/h_0 = 0.2$ over the conical island.

In numerical simulations, we consider breaking solitary waves with $H/h_0 = 0.28$ and $H/h_0 = 0.30$, as well as a non-breaking solitary wave with $H/h_0 = 0.0185$, with a water depth of $h_0 = 1$ m. For numerical discretization, we utilize the grid resolution of $\Delta x/h_0 = 0.04$. The initial numerical configuration of Example 4 is illustrated in Fig. 13. It should be noted that the figure is not to scale.

Figs. 14 and 15 show the temporal evolution of breaking solitary waves with $H/h_0 = 0.28$ and $H/h_0 = 0.30$ over the mild slope at different times $t^* = t\sqrt{g/h_0}$, respectively. Before reaching the shoreline, the front of the computed solitary wave becomes steeper than the experimental wave. This discrepancy arises from the limitation of the nonlinear SWEs, which lack a high-order dispersive term. These results have commonly been found in other studies solving the SWEs [e.g.35,36]. The laboratory wave was observed to break at approximately $t^* = 20$ as it traveled over the slope, past a region of constant water depth. However, the modeled wave acts as a bore rather than the breaking wave, propagating faster than the laboratory wave. These results have been commonly found in previous studies of breaking solitary waves [e.g.35,37]. At $t^* = 45$, the wave reaches its maximum run-up height, after which it starts to recede. Despite the limitations of the SWEs, there was a good agreement between the numerical solutions and laboratory measurements throughout the solitary wave run-up and rundown processes. Fig. 11 displays the initial water surface level along with the bottom topography.

Fig. 16 shows the temporal evolution of the non-breaking solitary wave with $H/h_0 = 0.0185$ on a sloping bed at various times. In contrast to the breaking solitary wave cases, the wave evolves due to the wave-shoaling process as it propagates over the slope, but no wave breaking occurs. It reaches its maximum run-up at $t^* = 55$, followed by the rundown process. The modeled wave aligns well

with the laboratory wave even before the run-up occurs due to the low H/h_0 condition. These results of breaking and non-breaking solitary wave propagations confirm the performance of the proposed scheme for simulating the evolution of the solitary waves over the slope.

3.5. Example 5—Two-dimensional solitary wave run-up on a conical island

As a final benchmark, a laboratory experiment of solitary wave propagation over a conical island, performed by Briggs et al. [38], is simulated. This case has been widely used for validations of numerical models [e.g. 19,22,39–48] even though the generated wave conditions are dispersive [e.g. 42,49]. The experiment domain consists of a channel with 25 m in length along the x -axis and 27.6 m in width along the y -axis, featuring a surface-piercing island with a height of 0.625 m. The bottom topography of this conical island is given by

$$B(x, y) = \begin{cases} 0.625, & r \leq 1.1, \\ 0.25(3.6 - r), & 1.1 < r \leq 3.6, \quad r := \sqrt{(x - 12.96)^2 + (y - 13.8)^2} \\ 0, & \text{otherwise,} \end{cases} \quad (50)$$

The initial conditions corresponding to the solitary wave of incident $H/h_0 = 0.1$ and $H/h_0 = 0.2$ are given by

$$\begin{aligned} \eta(x, y) &= H \operatorname{sech}^2 \left[\sqrt{\frac{3H}{4h_0^3}} \left\{ (x - x_0) \cos \theta + (y - y_0) \sin \theta \right\} \right], \\ u(x) &= \sqrt{\frac{3H}{4h_0^3}} \left(\frac{\eta}{\eta + h_0} \right) \cos \theta, \quad v(x) = \sqrt{\frac{3H}{4h_0^3}} \left(\frac{\eta}{\eta + h_0} \right) \sin \theta, \end{aligned} \quad (51)$$

where $h_0 = 0.32$ m and $(x_0, y_0) = (5, 13.8)$ is the center location of the solitary wave at $t = 0$, and $\theta = 0$ is the wave direction. The solid wall boundary conditions are imposed along the x -directions whereas the open boundary conditions are imposed along the y -directions. The Manning coefficient $n = 0.016$ is adopted to account for bed friction effects. A grid size of 0.15 m is used and the total simulation time is 20 s. Fig. 17 displays the initial water surface level along with the bottom topography.

Figs. 18 and 19 present a comparison between the computed and observed time evolution of water surface elevation over 20 seconds at gauges 1, 6, and 9 in front of the island, gauge 16 on the side, and gauge 22 behind the island (refer to Briggs et al. [38] for gauge ID and locations). These gauges provide sufficient coverage of the representative wave conditions in the experiment [35]. A strong correlation is noted between the numerical solutions and the laboratory experimental data, with accurate predictions of both the leading wave height and the arrival time of the leading waves. Comparable numerical results are found in previous studies [e.g. 19,22,35]. In the case of the breaking solitary wave with $H/h_0 = 0.2$, the results indicate a steeper front of the leading waves with earlier arrival time before wave run-up (see Fig. 19 (a) and (b)), attributed to the absence of a high-order dispersive term. Other studies [35,39,40] have also produced similar numerical solutions using their models. In summary, the two different solitary wave propagation cases over the conical island confirm the performance of the proposed scheme in simulating the evolution of solitary waves over two-dimensional bathymetry.

4. Conclusions

In this paper, we introduce a novel numerical scheme for the two-dimensional reconstruction and correction of water surface elevation at wet/dry fronts for the SWEs. Building on a one-dimensional positivity-preserving and well-balanced reconstruction scheme, we extend it into two-dimensional spaces, paying special attention to partially flooded cells with different bottom gradients along the x and y directions. A draining timestep for two-dimensional space is introduced to ensure the non-negativity of the computed water depth across the entire domain. We prove the positivity-preserving and well-balancing properties of the proposed scheme in the presence of wet/dry fronts. The proposed scheme has been validated against several numerical experiments with either analytical solutions or experimental data. These examples confirm that the proposed scheme accurately captures wet/dry interfaces in a non-oscillatory manner and minimizes numerical fluctuations during the wave run-up and rundown processes.

CRedit authorship contribution statement

Sooncheol Hwang: Methodology, Software, Formal analysis, Validation, Visualization, Writing - original draft; **Patrick J. Lynett:** Conceptualization, Methodology, Writing - review & editing; **Sangyoung Son:** Supervision, Funding acquisition, Conceptualization, Methodology, Investigation, Writing - review & editing.

Data availability

Data will be made available on request.

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